

Discussion Week 1

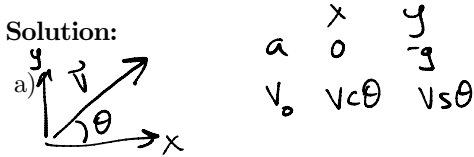
September 2026

1 Projectile Motion on a Flat Plane

Suppose a projectile is launched at some angle above horizontal (we'll call it θ , and keep it arbitrary), and at some initial launch speed v .

- Find expressions for the projectile's flight time, maximum height, and horizontal distance covered in flight in terms of the given quantities and physical constants.
- Using your results, find the launch angle θ_{max} that maximizes the distance travelled (no calculus allowed!!).
- Confirm your last result using the standard calculus procedure.

Solution:



$$t: y(t) = -\frac{1}{2}gt^2 + v \sin \theta t \rightarrow y(t_F) = 0 \rightarrow t_F = 0 (\text{launch}), \frac{2v \sin \theta}{g}$$

$$\text{OR } v_y(t_F) = 0 \rightarrow v(t) = v \sin \theta - gt \rightarrow t_F = \frac{2v \sin \theta}{g}$$

$$H_{max}: y(t_F) = y(t_{v_y=0}) = -\frac{1}{2}g\left(\frac{v \sin \theta}{g}\right)^2 + \frac{(v \sin \theta)^2}{g} = \frac{v^2 \sin^2 \theta}{2g}$$

Note: this 2nd method gives time of maximum height, that both work up to the factor of 2 is a feature of the symmetric surface (i.e. don't always rely on it)

$$\text{b) Dist: } x(t_F) = v \cos \theta \left(\frac{2v \sin \theta}{g}\right) = \frac{v^2}{g} (2 \sin \theta \cos \theta) = \frac{v^2}{g} \sin 2\theta$$

Distance has a sine function, max. value is 1 (when $\arg = \frac{\pi}{2} + 2n\pi, n \in \mathbb{Z}$)
so we want $2\theta_{max} = 90^\circ$ or $\theta_{max} = 45^\circ$

$$\text{c) } \frac{dD}{d\theta} = \frac{v^2}{g} 2 \cos 2\theta \rightarrow \text{set to } 0, \text{ find } 2\theta_{max} = \text{half integer multiples of } \pi$$

Only $\theta_{max} = 45^\circ$ physically relevant

2 Motion of an Electron in an Electric Field

Electromagnetic waves contain oscillating electric and magnetic fields, and we will consider the motion of a charge $-e$ (this is the standard notation for electric charge, where e is the "elementary charge" and like g is positive by convention. If curious, look up its value or ask your TA, otherwise you'll become plenty familiar with it in Physics 248!) in the electric field of such a wave taking the form

$$\vec{E} = \vec{E}_0 \sin \omega t \quad (1)$$

where $\vec{E}_0$ is some vector of fixed magnitude and direction. The force experienced by the electron is given by

$$\vec{F} = -e\vec{E} \quad (2)$$

After choosing a suitably convenient coordinate system, find expressions for the electron's velocity and position, assuming it was initially at rest. Interpret your results and see if you can explain any unexpected features.

Solution:

$\vec{E}_0$ is a constant vector, so we choose $\vec{E}_0 = E_0 \hat{x}$ and solve a 1D problem

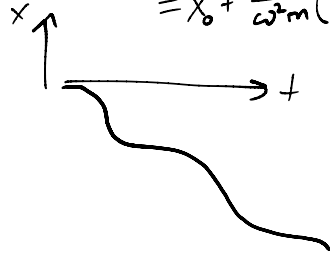
$$a_x = -\frac{eE_0}{m} \sin \omega t$$

$$v_x = v_{0x} + \int_0^t a_x(t') dt' = 0 + \frac{eE_0}{\omega m} \cos \omega t \Big|_0^t = \frac{eE_0}{\omega m} (\cos \omega t - 1)$$

$$x(t) = x_0 + \int_0^t v_x(t') dt' = x_0 + \frac{eE_0}{\omega^2 m} \left[\frac{\sin \omega t'}{\omega} - t' \right]_0^t$$

$$= x_0 + \frac{eE_0}{\omega^2 m} (\sin \omega t - \omega t)$$

Sinusoidal force leads to linear motion!



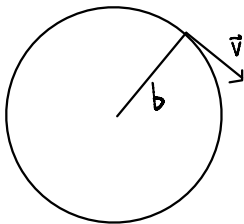
3 Polar Vectors

This exercise is meant to highlight the importance of choosing an appropriate (and convenient) coordinate system.

- Consider first a particle on a fixed length string (length b). As it spins, find an expression for its velocity in polar coordinates
- Suppose the string breaks right at the top of the circle. Write the particle's velocity after the string breaks, also in polar coordinates.

Solution:

a)

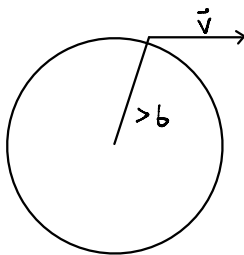


Polar unit vectors: $\hat{r}$ outward from origin
 $\hat{\theta}$ counterclockwise around origin } still $\perp$, like Cartesian vectors

$$\begin{aligned} \vec{V} &= v_r \hat{r} + v_\theta \hat{\theta} \\ &= 0 \hat{r} - \omega b \hat{\theta} = -\omega b \hat{\theta} \end{aligned}$$

$\uparrow$ constant radius $\nwarrow$ clockwise

b)



With no remaining forces, velocity vector is now constant, its magnitude is still ωb as before, but we need to be more careful resolving it along our unit vectors

using trig... $v_r = v \cos \theta$
 $v_\theta = -v \sin \theta$

$$\vec{V} = \omega b (\cos \theta \hat{r} - \sin \theta \hat{\theta})$$

Already this is obviously less convenient than for the rotating case, but it gets worse. Now, it is parametrized in terms of θ , and that angle no longer changes at a constant rate (it decreases at an ever shrinking rate, see the general velocity vector with v_r and v_θ above, and consider how to force its magnitude to stay constant)